

# BIS 629 Lecture 1: Theory of hypothesis testing

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# 1. Specification of the hypothesis:

## a. The standard hypothesis

The hypothesis test:  $H_0 : \beta^* = 0$  vs.  $H_A : \beta^* \neq 0$

- $\beta^*$  is the parameter of interest, also called the effect estimate, the treatment effect, the intervention effect
- This hypothesis test is a two-sided test, because under the alternative, we take an agnostic view and allow for  $\beta^*$  to be either positive or negative.

The two-sided test generally has less power than the one-sided test,  
 $H_0 : \beta^* \leq 0$  vs.  $H_A : \beta^* > 0$

- In implementation science, and biomedical research in general, one-sided tests are rarely used.

# 1. Specification of the hypothesis:

## a. The standard hypothesis

### Why two-sided tests?

There have been surprises, where a treatment believed to be beneficial turned out to be harmful

There are many examples, but here is one:

The Beta-Carotene and Retinol Efficacy Trial (CARET) tested the effect of daily vitamins A and C on the incidence of lung cancer in 18,314 high risk participants. CARET was stopped ahead of schedule because participants who were randomly assigned to the intervention had a 28% increase in lung cancer incidence, a 17% increase in death and higher cardiovascular disease mortality compared with the placebo group.

Reference: Effects of a combination of beta carotene and vitamin A on lung cancer and cardiovascular disease. Omenn GS, Goodman GE, Thornquist MD, Balmes J, Cullen MR, Glass A, Keogh JP, Meyskens FL, Valanis B, Williams JH, Barnhart S, Hammar S. N Engl J Med. 1996 May 2;334(18):1150-5.

Unless otherwise noted, in this course, we will be considering two-sided tests.

# Example

We compare the difference between sample means in an intervention and treatment group. Here,  $\beta^*$ , the effect estimate, is  $\mu_2 - \mu_1$ , where  $\mu_1$  is the mean of the outcome in the control group and  $\mu_2$  is the mean of the outcome in the intervention group.

Under the two-sided test with Type I error rate = 5%, we reject  $H_0$  when the test statistic, here the Z-score,

$$Z = \frac{\hat{\mu}_2 - \hat{\mu}_1}{\sqrt{\widehat{Var}(\hat{\mu}_2 - \hat{\mu}_1)}}$$

is  $> 1.96$  or  $< -1.96$ , the 2.5th and 97.5th percentiles of the standard normal distribution,  $N(0, 1)$ .

This means, that, when the null hypothesis is true, it will be rejected incorrectly 5% of the time. Type I error rates of 5% are nearly universal in implementation science and other biomedical research.



## Example: Familia salama

Goal: Increase the number of antenatal care (ANC) visits during pregnancy to the World Health Organization (WHO) recommended standard of 4 or more during the pregnancy

$\beta^*$  is the difference between the number of ANC visits among women in the intervention wards compared to the number of ANC visits among women in the control wards –

- estimated with continuous, non-normally distributed outcome data

$$H_0 : \beta^* = 0 \text{ vs. } H_A : \beta^* \neq 0$$

Let  $Y_{ij} = 1$  if woman  $i$  in ward  $j$  had 4 or more ANC visits, 0 otherwise.

Now  $\beta^*$  is the difference in the proportion of participants with  $\geq 4$  ANC visits among women in the intervention wards vs. the control wards –  $H_0 : \beta^* = 0$  vs.  $H_A : \beta^* \neq 0$

- estimated with binary data, identity link function – difference between two binomial proportions

# Example: Familia salama

Or  $\beta^*$  is the ratio:

$\beta^*$  is the ratio of the proportions  $\geq 4$  ANC visits among women in the intervention wards vs. the control arms –

- estimated with binary data, log or logistic link function – ratio or odds ratio between two binomial proportions:

$$H_0 : \beta^* = 1 \text{ vs. } H_A : \beta^* \neq 1$$

$\equiv$

$$H_0 : \frac{p_2}{p_1} = 1 \text{ vs. } H_A : \frac{p_2}{p_1} \neq 1$$

That is,  $\beta^*$  is defined as  $\frac{p_2}{p_1}$ .

Note that by simple re-arrangement,

$$H_0 : \frac{p_2}{p_1} = 1 \equiv H_0 : p_2 = p_1 \equiv H_0 : p_2 - p_1 = 0$$

$$\text{And also } H_0 : \frac{\frac{p_2}{1-p_2}}{\frac{p_1}{1-p_1}}$$

# 1. Specification of the hypothesis:

## b. The non-inferiority hypothesis

In implementation science, we may also be interested in the non-inferiority hypothesis. In the related discipline of comparative effectiveness research, the non-inferiority hypothesis is of considerable importance.

For example, HLB-Simple trial, Uganda, HIV+ patients under HIV care – 3 arms

- 1 Standard of care for hypertension control
- 2 HTN-Basic: receive free and consistent access to diagnostic equipment and evidence-based antihypertensive drugs
- 3 HTN-Plus: HTN-Basic plus a multi-component implementation strategy to improve uptake and adherence to BP treatments, contextually adapted to these clinics, using a human-centered design approach to adapt differentiated service delivery models, use of hypertensive PLHIV peer champions, and methods of blood pressure monitoring to address specific barriers to, and facilitators of, blood pressure care.

- The primary effectiveness outcome will be % of patients with hypertension diagnosis who are controlled ( $< 140\text{mmHg}$  systolic)— estimated as a binomial proportion
- A question of interest is whether HTN-Plus improves hypertension control compared to HTN-BASIC
- Now, this question can be formulated as a standard null hypothesis as above, where  $\beta^*$  = the difference in proportions under control in each arm, and the null hypothesis is that HTN-BASIC,  $p_1$ , equals the proportion under control in HTN-PLUS,  $p_2$ .
- Alternatively, we can investigate whether HTN-BASIC is not inferior to HTN-PLUS, vs. the alternative that it is inferior, with non-inferiority specified by a parameter  $\delta$ . For example, it might be reasonable to test the hypothesis that  

$$H_0 : p_2 - p_1 \leq \delta \text{ (new intervention is not inferior)} \text{ vs.}$$

$$H_A : p_2 - p_1 > \delta \text{ (new intervention is inferior, with inferiority defined as } p_2 \text{ being greater than } p_1 \text{ by a margin } \delta)$$

Then,  $Z = \frac{(\hat{p}_2 - \hat{p}_1 - \delta)}{\sqrt{\hat{\text{Var}}(\hat{p}_1) + \hat{\text{Var}}(\hat{p}_2)}}$ , and  $p = 1 - \Phi(Z)$ , where  $\Phi(\cdot)$  is the cumulative standard normal

# Non-inferiority tests

- Perhaps HTN-PLUS is a little better, but it may not be worth the time effort, expense unless it improves hypertension control compared to HTN-BASIC by, for example,  $\delta = 5$  percentage points or more.
- Another way to write a non-inferiority hypothesis uses one-sided confidence intervals.
- Here, we test  $H_0 : p_2 - p_1 \leq \delta$  through the one-sided confidence interval  $\hat{p}_2 - \hat{p}_1 + 1.645\sqrt{\widehat{Var}(\hat{p}_1) + \widehat{Var}(\hat{p}_2)} = CI$ , and reject  $H_0$  if  $CI > \delta$ , concluding that  $p_1$  is inferior, otherwise, not.
- In clinical research, non-inferiority tests are of interest as well
- Here the concern is that a new intervention might improve a primary outcome that is beneficial on a primary outcome, but may also have adverse side effects.
- Then, we may wish to test the null hypothesis that the new intervention is 'worse' than the control with respect to this adverse outcome, vs. the alternative that it is not worse (non-inferior)

# Non-inferiority tests

- For example, in Familia Salama, Option B may be more effective in preventing HIV transmission from mother to child, but there is concern may also cause birth defects.
- Let  $p_1$  be the rate of birth defects among the babies of mothers who received option A, and  $p_2$  be the same for Option B. We would not like the birth defect rate to be more than  $\delta = 0.5\%$  percentage points higher than Option A in the Option B arm. Thus,  $\delta = 0.05 \times p_1$
- Hence,  $H_0 : p_2 - p_1 \leq \delta$  vs.  $H_A : p_2 - p_1 > \delta$
- Alternatively, one could specify the null hypothesis such that we reject  $H_0$  when the upper bound of the 95% confidence interval for  $\hat{p}_2 - \hat{p}_1$  is  $> \delta$ , that is, we reject  $H_0$  when  $\hat{p}_2 - \hat{p}_1 + 1.645\sqrt{\text{Var}(\hat{p}_1) + \text{Var}(\hat{p}_2)} > \delta$

# Equivalence tests

- Equivalence tests are one-sided tests against a known standard. For example, if we knew that the birth defect rate in the general population was  $\delta = 0.01\%$ , we might wish to assess the consistency of the data from Option B with population birth defect rates
- That is,  $H_0 : p_2 \geq \delta$  vs.  $p_2 < \delta$ , where under  $H_0$  the intervention is not equivalent to the national standard, and under  $H_A$  it is
- Alternatively, one could specify the null hypothesis such that if the upper bound of the 95% confidence interval for  $p_2$  is  $< \delta$ , that is, if  $\hat{p}_2 + 1.645\sqrt{\widehat{Var}(\hat{p}_2)} < \delta$ , we reject  $H_0$  and conclude that the intervention is no worse than the background standard.

Any hypothesis can be specified. Then it remains to derive, given the study design and available data, how to construct the test statistic and assess its significance.

# Construction of the test statistic

- We need to assess the consistency of the data generated in the study with the null hypothesis.
  - if it's not consistent with the null hypothesis we reject it and conclude the alternative hypothesis is more consistent with the data, that is, the test was 'significant'.
- To do this we construct a test statistic.
  - The test statistic is a function of the data with a known asymptotic distribution under the null hypothesis and under the alternative.
- We will consider asymptotic tests only in this course because implementation science tends to work with large studies. Situations where asymptotic results may be less reliable will be discussed.

The test statistic,  $Z^2(\mathbf{Y}, \mathbf{X}; \beta)$  is a function of the data,  $\{\mathbf{Y}, \mathbf{X}\}$ , where

- $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)^T$ , the outcomes for participants  $j = 1, \dots, n$  in the study, with  $n$  the total sample size, and
- $\mathbf{X} = (X_1, X_2, \dots, X_n)^T$  is the intervention/treatment status for the study participants.
- $\beta$  is a vector of parameters, where  $\beta^*$  is the parameter of interest in  $\beta$ . The number of parameters is  $q$ .
- Further notation will be introduced as needed as the course progresses. See the end of lecture notes for a dictionary of notation.

# Classic test statistics

The two classic test statistics are

- the test for the difference between two sample means, and
- the test for the difference between two binomial proportions
- Most other test statistics are derivatives of these two

The **test for the difference between two sample means** has the following form

$$Z^2 = \frac{(\hat{\mu}_2 - \hat{\mu}_1)^2}{\widehat{Var}(\hat{\mu}_2 - \hat{\mu}_1)},$$

where

$\hat{\mu}_1 = \sum_{j=1}^{n_1} Y_j / n_1$ , and  $Y_j$ ,  $j = 1, \dots, n_1$  are the continuous outcome data for Arm 1 of the study, and similarly for  $\hat{\mu}_2$ ,  $j = n_1 + 1, \dots, n_1 + n_2$

## Test for the difference between two sample means

- Asymptotically, that is, as  $n_1 \rightarrow \infty$  and  $n_2 \rightarrow \infty$ , under the central limit theorem, under  $H_0$ ,

$$Z^2 \xrightarrow{d} \chi_1^2 \quad (\text{central } \chi^2(1)),$$

a central  $\chi^2$  distribution with 1 degree of freedom.

And under  $H_A$ ,

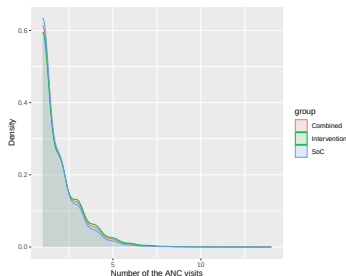
$$Z^2 \xrightarrow{d} \chi_1^2(\lambda),$$

a non-central central  $\chi^2$  distribution with 1 degree of freedom and non-central parameter,  $\lambda$ , where  $\lambda = \frac{\beta^{*2}}{\text{Var}(\hat{\mu}_2 - \hat{\mu}_1)} | (\beta = \beta^*)$

- These asymptotic results apply regardless of the distribution of  $\mathbf{Y} = (\mathbf{Y}_1^T, \mathbf{Y}_2^T)$ , where  $\mathbf{Y}_1 = (Y_1, \dots, Y_{n_1})^T$  and  $\mathbf{Y}_2 = (Y_{n_1+1}, \dots, Y_{n_1+n_2})^T$ .
- These results will also apply if  $\mathbf{Y}_1 \sim N(\mu_1, \sigma_1^2)$  and  $\mathbf{Y}_2 \sim N(\mu_2, \sigma_2^2)$  exactly, that is, if the outcome data from both groups are normally distributed.

# Example: Familia Salama, Dar es Salaam, Tanzania

| Group                                              | Intervention          | SoC                  | Total                 |
|----------------------------------------------------|-----------------------|----------------------|-----------------------|
| Sample size                                        | N=134,938             | N=108,980            | N=243,918             |
| Proportion of $\geq 4$ ANC visits                  | 12,560/134,938 (9.3%) | 7,155/108,980 (6.6%) | 19,715/243,918 (8.1%) |
| ICC (95% CI) for Proportion of $\geq 4$ ANC visits | 0.05 (0.03, 0.07)     | 0.07 (0.05, 0.11)    | 0.05 (0.03, 0.07)     |
| Mean (SD) for number of ANC visits                 | 1.79 (1.21)           | 1.67 (1.06)          | 1.74 (1.15)           |
| ICC (95% CI) for number of ANC visits              | 0.07 (0.05, 0.10)     | 0.10 (0.07, 0.15)    | 0.06 (0.04, 0.08)     |



$$Z^2 = \frac{(1.79 - 1.67)^2}{1.21^2/134938 + 1.06^2/108980} = 680.52$$

$$p\text{-value} = 1 - Pr(Z^2 \leq \chi^2_{1,\alpha} | H_0) = 0$$

Why is this p-value wrong? We will correct this later.

# Asymptotic theory for test statistics

- Let  $\beta$  is the parameter vector indexing the function of the data upon which the test statistic is based, where usually in implementation science although not always (e.g. the interrupted time series design), the parameter of interest,  $\beta^*$ , is a scalar element from this vector, with the remaining elements being nuisance parameters.
- For a given scenario, the challenge is to find a test statistic must have a known probability distribution under the null and its alternative.
- When considering asymptotic test statistics, we need to identify a test statistic that converges in law to a standard  $N(0, 1)$  under  $H_0$ . Then, the square of this test statistic follows the central and non-central  $\chi^2$  distribution under  $H_0$  and  $H_A$  respectively, providing two-sided critical values in a straightforward manner.

### 3 standard approaches

- Wald test
- Likelihood ratio test
- Score test

We will consider  $H_0 : \beta^* = 0$  vs.  $H_A : \beta^* \neq 0$  unless otherwise specified.

Let the log-likelihood function be

$$\sum_{j=1}^n \log f(Y_j | X_j; \beta),$$

where  $f(Y_j | X_j; \beta)$ , the probability density model that is assumed to have generated the data,  $(Y_j, X_j)$ , has a form such that it includes  $\beta^*$ , the parameter of interest, in  $\beta$ .

# Likelihood-based approaches for constructing test statistics

- The maximum likelihood estimator, MLE,  $\hat{\beta}$ , is

$$\arg \max_{\beta} \sum_{j=1}^n \log f(Y_j | X_j; \beta),$$

- which is equivalent to the solution with respect to  $\beta$  of

$$\frac{\partial \sum_{j=1}^n \log f(Y_j | X_j; \beta)}{\partial \beta} = \mathbf{0}$$

- note that there are  $q$  score equations that must be simultaneously solved

# Likelihood-based approaches for constructing test statistics

- The asymptotic variance of  $\hat{\beta}^*$ , is the  $r$ - $r$  element of the variance-covariance matrix

$$\text{Var}(\hat{\beta}) = \left[ -E_{(\mathbf{Y}|\mathbf{X})} \left\{ \frac{\partial^2 \sum_{j=1}^n \log f(Y_j | X_j; \beta)}{\partial \beta \partial \beta^T} \right\} \right]^{-1}.$$

Note that this variance is not a function of  $\mathbf{Y}$  and  $\beta^*$  is the  $r$ th element of  $\beta$ .

- Sometimes people write **AVar** instead of **Var** to indicate the asymptotic variance. Because we only consider asymptotic variances in this class, we will write **Var**.

# 1. Wald test

$$Z^2 = \frac{(\hat{\beta}^*)^2}{\text{Var}(\hat{\beta}^*)}$$

where  $\text{Var}(\hat{\beta}^*)$  was defined above.

This test is asymptotically equivalent to

$$Z^2 = \frac{(\hat{\beta}^*)^2}{\widehat{\text{Var}(\hat{\beta}^*)}} \Big|_{\beta^* = \hat{\beta}^*}.$$

- *Asymptotically equivalent* means that the statistics converge in probability to the same value and have the same asymptotic normal distribution with the same variance. They are not the same functions of the data, that is, they are not, equal.

# 1. Wald test

- Under  $H_0$ ,  $Z^2 \xrightarrow{d} \chi_1^2$ .
- Under  $H_A$ ,  $Z^2 \xrightarrow{d} \chi_1^2(\lambda)$ , where  $\lambda$ , the non-centrality parameter, is
$$\lambda = \frac{(\beta_{HA}^*)^2}{\text{Var}(\hat{\beta}^*)}$$
- Example: Logistic regression
  - Here,  $f(Y_j|X_j) = \frac{\exp(\beta_0 + \beta^* X_j)^{Y_j}}{1 + \exp(\beta_0 + \beta^* X_j)}$  and the probability of  $P(Y_j = 1|X_j; \beta)$ :
$$p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)},$$
  - The log-likelihood for the study with data  $\{\mathbf{Y}, \mathbf{X}\}$  of sample size  $n = n_1 + n_2$ ,

$$\begin{aligned} L(\beta) &= \sum_{j=1}^n \log f(Y_j|X_j; \beta) \\ &= \sum_{j=1}^n Y_j(\beta_0 + \beta^* X_j) - \log\{1 + \exp(\beta_0 + \beta^* X_j)\} \end{aligned} \quad (1)$$

# 1. Wald test

- The asymptotic variance of  $\hat{\beta}^*$ , is the 2 – 2 element of the variance-covariance matrix

$$Var(\hat{\beta}) = \left[ -E_{(\mathbf{Y}|\mathbf{X})} \left\{ \frac{\partial^2 \sum_{j=1}^n \log f(Y_j|X_j; \beta)}{\partial \beta \partial \beta^T} \right\} \right]^{-1}$$

# 1. Wald test

where

$$\sum_{j=1}^n \frac{\partial \log f(Y_j | X_j; \beta)}{\partial \beta} = \sum_{i=1}^n \begin{bmatrix} 1 \\ X_j \end{bmatrix} \left\{ Y_j - \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)} \right\} = \begin{bmatrix} \sum_{j=1}^n \left\{ Y_j - \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)} \right\} \\ \sum_{j=1}^n X_j \left\{ Y_j - \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)} \right\} \end{bmatrix}$$

and  $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$ . Then,

$$\frac{\partial^2 \sum_{i=1}^n \log f(Y_j | X_j; \beta)}{\partial \beta \partial \beta^T} = - \sum_{j=1}^n \begin{bmatrix} 1 \\ X_j \end{bmatrix} p_j(1 - p_j) \begin{bmatrix} 1 & X_j \end{bmatrix} = - \sum_{i=1}^n \begin{bmatrix} p_j(1 - p_j) & X_j p_j(1 - p_j) \\ X_j p_j(1 - p_j) & X_j^2 p_j(1 - p_j) \end{bmatrix}$$

Since the above expression is not a function of  $Y_i$ ,

$$-E_{(Y|X)} \left\{ \frac{\partial^2 \sum_{j=1}^n \log f(Y_j | X_j; \beta)}{\partial \beta \partial \beta^T} \right\} = - \frac{\partial^2 \sum_{j=1}^n \log f(Y_j | X_j; \beta)}{\partial \beta \partial \beta^T} = \text{Var}(\hat{\beta})$$

# 1. Wald test

- After simplification, we have

$$\text{Var}(\hat{\beta}^*) = \frac{\sum_{j=1}^n p_j(1 - p_j)}{\left\{ \sum_{i=1}^n p_j(1 - p_j) \right\} \left\{ \sum_{j=1}^n X_j^2 p_j(1 - p_j) \right\} - \left\{ \sum_{i=1}^n X_j p_j(1 - p_j) \right\}^2}$$

- And

$$\widehat{\text{Var}}(\hat{\beta}^*) = \frac{\sum_{j=1}^n \hat{p}_j(1 - \hat{p}_j)}{\left\{ \sum_{j=1}^n \hat{p}_j(1 - \hat{p}_j) \right\} \left\{ \sum_{j=1}^n X_j^2 \hat{p}_j(1 - \hat{p}_j) \right\} - \left\{ \sum_{j=1}^n X_j \hat{p}_j(1 - \hat{p}_j) \right\}^2} \quad (2)$$

$$\text{where } \hat{p}_j = \frac{\exp(\hat{\beta}_0 + \hat{\beta}^* X_j)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}^* X_j)}.$$

## Example: HPTN 037

- The HIV Prevention Trials Network (HPTN) 037 Study evaluated a peer-driven intervention to promote HIV risk reduction among people who inject drugs (PWID) and their injection drug and sexual network members in Philadelphia, PA, US.
- PWID volunteered as indexes, served the initial point of contact into their HIV risk network, and recruited other immediate HIV risk network members, which formed non-overlapping egocentric networks around each index member.
- Eligible indexes and their risk networks were randomized to the experimental intervention or control. Follow-up visits occurred at six-month intervals following randomization for up to 30 months.
- 232 indexes and 419 network members were examined to assess changes in injection risk behaviors.

## Example: HPTN 037

- In this example, we will take the primary outcome to be the cumulative 30-month risk injection behavior, “**shared cotton**”, which is an indicator variable for sharing needle/syringe “works”.
- $Y_j = 1$  for participants reporting the risk injection behavior over follow-up,  $Y_j = 0$  for otherwise.
- $X_j = 1$  for the participant in the intervention group,  $X_j = 0$  for the participant is in control group.
- total  $n = 696$  participants
- Logistic regression is used to estimated the overall effect of the intervention:

$$f(Y_j|X_j) = \frac{\exp(\beta_0 + \beta^* X_j)^{Y_j}}{1 + \exp(\beta_0 + \beta^* X_j)}$$

- We are interested in testing  $H_0 : \beta^* = 0$  vs.  $H_0 : \beta^* \neq 0$ .

## Example: HPTN 037

- We first estimate the parameters using GEE:

```
fit1 <-geeglm( cotton ~ Treat , data = base_hptn_30 ,  
              id=uid , family = binomial(link="logit" ) )
```

and we obtain that  $\hat{\beta} = (-0.022, -0.314)$ ,

- Then  $\widehat{Var}(\hat{\beta}^*)$  can be calculated by (2), which is 0.0234 (SE = 0.153);
- $Z^2 = \frac{(\hat{\beta}^*)^2}{\widehat{Var}(\hat{\beta}^*)} = 4.228$ ;
- Given  $\alpha = 0.05$ , p-value =  $P(Z^2 > 4.228) = 0.0397 < 0.05$
- Thus, we reject the  $H_0$  given 5% significance level.
- The result can be checked by using

```
> summary(fit1)  
  
Call:  
geeglm(formula = cotton ~ Treat, family = binomial(link = "logit"),  
       data = base_hptn_30, id = uid)  
  
Coefficients:  
              Estimate Std.err Wald Pr(>|w|)  
(Intercept)  -0.0222   0.1054  0.04    0.83  
Treat        -0.3142   0.1528  4.23    0.04 *  
---  
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## 2. Score test

- The first derivative of the log likelihood is often referred to as the score function or score statistic, and denoted  $U_j$ .

$$U_j = U_j(Y_j, X_j; \beta) = \frac{\partial \log f(Y_j|X_j; \beta)}{\partial \beta} \text{ and } \mathbf{U} = \sum_{j=1}^n U_j(Y_j, X_j; \beta).$$

- Note that by definition of the MLE,  $\sum_{j=1}^n U_j(\hat{\beta}) = \mathbf{0}$  since the MLE is the solution to this equation, where, here,  $\dim(U_j) = 2 \times 1$  corresponding to  $(\beta_0, \beta^*)^T$ .
- In addition, for GLMs such as logistic and linear regression,  $U_j$  has the form  $w_j(X_j; \beta)[Y_j - E(Y_j|X_j; \beta)]$ , guaranteeing that  $E_{Y|X}[w_j(X_j; \beta)(Y_j - E(Y_j|X_j; \beta))] =$   
 $w_j(X_j; \beta)[E_{Y|X}(Y_j|X_j; \beta) - E(Y_j|X_j; \beta)] = 0$
- In general, if  $\hat{\beta}$  solves unbiased estimating equations such as the above,  $\hat{\beta}$  is consistent for  $\beta$  (Rao 1965\*, Huber 1967†, Liang and Zeger 1986‡)

\* Rao, Calyampudi Radhakrishna, et al. Linear statistical inference and its applications. Vol. 2. New York: Wiley, 1973.

† Huber, Peter J. "UNDER NONSTANDARD CONDITIONS." Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability 1 (1967): 21-37.

‡ Liang, Kung-Yee, and Scott L. Zeger. "Longitudinal data analysis using generalized linear models." Biometrika 73.1 (1986): 13-22.

## 2. Score test

- It has been shown that

$$\text{Var}(U_j) = -E_{Y_j|X_j} \left\{ \frac{\partial^2 f_j(Y_j, X_j; \beta)}{\partial \beta \partial \beta^T} \right\} = E_{Y_j|X_j}(U_j U_j^T),$$

and we denoted  $\text{Var}(\mathbf{U}) = \sum_{j=1}^n \text{Var}(U_j)$  as  $\mathbf{I}$ . Note that either expression can be used to estimate the variance.

- Thus, the score test statistic for  $H_0 : \beta^* = 0$  vs.  $H_A : \beta^* \neq 0$  takes the value of  $\mathbf{U}^T \mathbf{I}^{-1} \mathbf{U}$  as  $Z^2 \xrightarrow{d} \chi_1^2$  under  $H_0$ , where

$$\mathbf{U}(\beta_{H_0}) = \sum_{j=1}^n U_j(X_j; \beta^* = 0, \beta_{-r}),$$

$\beta_{-r}$  is the  $(r-1) \times 1$  vector of  $\beta$  where  $\beta^*$ , the  $r$ th element, is deleted. And

$$\mathbf{I}(\beta_{H_0}) = - \sum_{j=1}^n E_{Y_j|X_j} \left[ \frac{\partial^2 \log f(Y_j|X_j; \beta^* = 0, \beta_{-r})}{\partial \beta \partial \beta^T} \right]$$

## Example: Logistic regression

- Recall that  $f(Y_j|X_j) = \frac{\exp(\beta_0 + \beta^* X_j)^{Y_j}}{1 + \exp(\beta_0 + \beta^* X_j)}$  and  $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$
- The log-likelihood for the study with data  $\{\mathbf{Y}, \mathbf{X}\}$  of sample size  $n$ ,

$$L(\beta) = \sum_{j=1}^n Y_j(\beta_0 + \beta^* X_j) - \log\{1 + \exp(\beta_0 + \beta^* X_j)\}$$

Then,

$$\begin{aligned} \mathbf{U}(\hat{\beta}_{H_0}) &= \sum_{j=1}^n \frac{\partial \log f(Y_j|X_j; \beta^* = 0, \hat{\beta}_{H_0})}{\partial \beta} = \sum_{i=1}^n \begin{bmatrix} 1 \\ X_j \end{bmatrix} \left\{ Y_j - \frac{\exp(\hat{\beta}_{H_0})}{1 + \exp(\hat{\beta}_{H_0})} \right\} \\ &= \begin{bmatrix} \sum_{j=1}^n \left\{ Y_j - \frac{\exp(\hat{\beta}_{H_0})}{1 + \exp(\hat{\beta}_{H_0})} \right\} \\ \sum_{j=1}^n X_j \left\{ Y_j - \frac{\exp(\hat{\beta}_{H_0})}{1 + \exp(\hat{\beta}_{H_0})} \right\} \end{bmatrix} \end{aligned} \quad (3)$$

## Example: Logistic regression

$$I(\hat{\beta}_{H_0}) = \sum_{j=1}^n \left[ \begin{array}{cc} \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \left\{ 1 - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \right\} & X_j \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \left\{ 1 - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \right\} \\ X_j \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \left\{ 1 - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \right\} & X_j^2 \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \left\{ 1 - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \right\} \end{array} \right]. \quad (4)$$

Then the score test statistic  $Z^2$  is

$$Z^2 = \frac{1}{Z_3} \left[ \left( \sum_{j=1}^n Z_{1j} \right) \left( \sum_{j=1}^n Z_{2j} \right)^2 - 2 \left( \sum_{j=1}^n Z_{2j} \right) \left( \sum_{j=1}^n X_j Z_{1j} \right) \left( \sum_{j=1}^n X_j Z_{2j} \right) + \left( \sum_{j=1}^n X_j^2 Z_{1j} \right) \left( \sum_{j=1}^n X_j Z_{2j} \right)^2 \right] \quad (5)$$

where  $Z_{1j} = \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \left\{ 1 - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \right\}$ ,  $Z_{2j} = Y_j - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})}$ , and

$$Z_3 = \left( \sum_{j=1}^n Z_{1j} \right) \left( \sum_{j=1}^n X_j^2 Z_{1j} \right) - \left( \sum_{j=1}^n X_j Z_{1j} \right)^2.$$

## Example: HPTN 037

- We first estimate the nuisance parameter  $\beta_0$  under  $H_0$  using GLM for the model under  $H_0$ :  $f(Y_j|X_j, H_0) = \frac{\exp(\beta_0)^{Y_j}}{1+\exp(\beta_0)}$ , we obtain that  $\hat{\beta}_{H_0} = \hat{\beta}_{H_0} = -0.173$ .
- We obtain  $\hat{\beta}_{0|H_0}$  is fitting the null model given the data:

```
> glm.fit0 <- glm(cotton~1, data = base_hptn_30, family = binomial(link="logit"))
> fit0=glm.fit0
> summary(fit0)
```

Call:

```
glm(formula = cotton ~ 1, family = binomial(link = "logit"),
    data = base_hptn_30)
```

Deviance Residuals:

| Min    | 1Q     | Median | 3Q    | Max   |
|--------|--------|--------|-------|-------|
| -1.105 | -1.105 | -1.105 | 1.252 | 1.252 |

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |
|-------------|----------|------------|---------|----------|
| (Intercept) | -0.17284 | 0.07609    | -2.271  | 0.0231 * |

---

## Example: HPTN 037

- Then  $\mathbf{U}(\hat{\beta}_{H_0})$  and  $\mathbf{I}(\hat{\beta}_{H_0})$  can be calculated using (3) and (4)
- $Z^2 = \mathbf{U}(\hat{\beta}_{H_0})^T \mathbf{I}(\hat{\beta}_{H_0})^{-1} \mathbf{U}(\hat{\beta}_{H_0}) = 4.237$
- Given  $\alpha = 0.05$ ,  $p\text{-value} = P(Z^2 > 4.237) = 0.040 < 0.05$
- Thus, we reject the  $H_0$  given 5% significance level.
- The result can be checked by using

```
> fit2 <- glm(cotton~ Treat, data = base_hptn_30, family = binomial(link="logit"))
> anova(fit2, test ="Rao")
Analysis of Deviance Table

Model: binomial, link: logit
Response: cotton

Terms added sequentially (first to last)
```

|       | Df | Deviance | Resid. Df | Resid. Dev | Rao  | Pr(>Chi) |
|-------|----|----------|-----------|------------|------|----------|
| NULL  |    |          | 695       | 960        |      |          |
| Treat | 1  | 4.24     | 694       | 955        | 4.24 | 0.04 *   |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### 3. The Likelihood ratio test (LRT)

- The test statistic  $Z^2$  is

$$\begin{aligned} Z^2 &= -2\{ \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \beta_{-r} = \hat{\beta}_{-r}) \\ &\quad - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = \hat{\beta}^*, \beta_{-r} = \hat{\beta}_{-r}) \} \\ &\xrightarrow{d} \chi_1^2 \end{aligned}$$

#### Example: HPTN 037

- Similar to the score test, we estimate the parameter using GLM method based on the model under  $H_0$ :  $f(Y_j|X_j; \beta_{H_0}) = \frac{\exp(\beta_0)^{Y_j}}{1 + \exp(\beta_0)}$ , we obtain  $\hat{\beta}_0 = \hat{\beta}_0 = -0.173$ ,

## Example: HPTN 037

- Then we estimate the parameter based on the model under  $H_A$ ,  
 $f(Y_j|X_j, H_A) = \frac{\exp(\beta_0 + \beta^* X_j)^{Y_j}}{1 + \exp(\beta_0 + \beta^* X_j)}$ , we obtain that  $\hat{\beta}_0 = (-0.022, 0.314)$ ,
- $Z^2 = -2\{\log f(\mathbf{Y}|\mathbf{X}, \hat{\beta}_0 = -0.173, \beta^* = 0) - \log f(\mathbf{Y}|\mathbf{X}, \hat{\beta}_0 = -0.022, \hat{\beta}^* = -0.314)\} = 4.242$
- Given  $\alpha = 0.05$ , p-value  $= P(Z^2 > 4.242) = 0.0394 < 0.05$
- Thus, we reject the  $H_0$ .
- The result can be checked by using

```
> anova(fit2, test = "LRT")
Analysis of Deviance Table

Model: binomial, link: logit

Response: cotton

Terms added sequentially (first to last)

      Df Deviance Resid. Df Resid. Dev Pr(>Chi)
NULL                                695      960
Treat  1      4.24      694      955    0.039 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

# Comparison of the three tests

Table 1: Hypothesis Tests Results of HPTN 037 data

| Tests      | Test Statistics ( $Z^2$ ) | P-value |
|------------|---------------------------|---------|
| Wald Test  | 4.228                     | 0.0397  |
| Score Test | 4.237                     | 0.0400  |
| LRT        | 4.242                     | 0.0394  |

- It has been found that the LRT has better finite sample properties than the score test, which has better finite sample properties than the Wald test. (Vaeth \*)

\* Vaeth M. On the use of Wald's test in exponential families. International Statistical Review 1985; 53:199-214

- For power calculations, the expected value of the LRT under the null and alternative is often difficult to calculate, and the Wald test is most commonly used, then the score test.

- Now that we have test statistics, we can consider power calculations.
- The power of the study is the probability that the null hypothesis will be rejected when it is false.
- The power of the study is calculated for a pre-specified value of  $\beta^*$ , the intervention effect, also known as the parameter of interest.
- The problem is set up so that the test statistic rejects the null  $1 - \alpha/2$  (for two-sided test) when the null hypothesis is true

- **Power( $\pi$ ):**

$\pi = Pr(Z^2 > c | H_A) = 1 - Pr(Z^2 \leq c | H_A) = 1 - Pr(Z_\lambda^2 \leq 3.84)$ , where  $c = 3.84$  is the critical value of a central  $\chi_1^2$  for  $\alpha = 5\%$ ,  $\lambda$  is the non-centrality parameter, a function of the alternative value of the effect estimate,  $\beta_{H_A}^*$ , of  $n$ , and other aspects of the design that determine  $Var_{H_A}(\hat{\beta}_{H_A}^*)$ .  $Z_\lambda^2$  is a random variable following the non-central  $\chi_1^2(\lambda)$  distribution with 1 degrees of freedom and non-centrality parameter  $\lambda$ .

# Sample size and Minimum detectable effect (MDE)

- **Sample size**,  $n$ , solves  $\pi = 1 - Pr(Z_{\lambda}^2 \leq 3.84)$ , where  $\lambda$  is a function of  $n$  through the variance, and  $\pi$  is pre-specified power, usually 80%.
- **Minimum detectable effect size (MDE)** Here, power and sample size are fixed.
  - This is often the case in grant proposals because the budget is limited.
  - Hence, with a given budget/sample size and the usual 80% power constraint, we want to know what our minimum detectable effect size is, that is,
    - Find  $\beta^*$  that solves  $\pi = 1 - Pr(Z_{\lambda}^2 \leq 3.84)$ , where  $\lambda$ , the non-centrality parameter is a function of  $n$  and  $\beta^*$  both in its numerator and in the denominator, at least for categorical/binomial/Bernoulli data.

# Finding the non-centrality parameter $\lambda$ for fixed power, $\pi$

- Given required power  $\pi$ , significance level  $\alpha$ , degrees of freedom of the  $\chi^2$  distribution, we can obtain the  $\lambda$  by solving

$$\pi = 1 - \int_0^{3.84} f(x; 1, \lambda) dx, \quad (6)$$

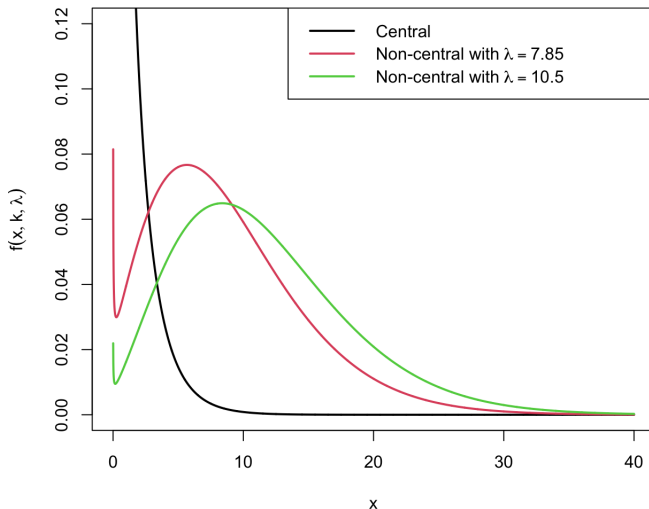
where  $f(x; 1, \lambda)$  is the probability density function of the noncentral chi-squared distribution with DF=1 and non-centrality parameter  $\lambda$ .

- For example, given  $\pi = 80\%$ ,  $\alpha = 5\%$  and DF=1, we have  $\lambda(\pi) = 7.85$ . When  $\pi = 90\%$ ,  $\lambda(\pi) = 10.5$

```
fx=function(x, lambda) dchisq(x, 1, ncp = lambda, log = FALSE)
power.eq1<-function(lambda,power=0.8) integrate(fx,0,3.84,lambda=lambda)$value - (1-power)
##### (1) lambda for 80% power
rootSolve::multiroot(power.eq1 , start =10, positive = TRUE, power=0.8)$root
# [1] 7.846775
##### (2) lambda for 90% power
rootSolve::multiroot(power.eq1 , start =10, positive = TRUE, power=0.9)$root
# [1] 10.5049
```

# Finding the non-centrality parameter $\lambda$ for fixed power, $\pi$

Probability density function of  $\chi_1^2$  with non centrality parameter  $\lambda$



# Three types of power analysis

- To find the power,  $\pi$ , for a study with sample size  $n$  assuming  $\beta = \beta^*$ , we calculate  $\pi = 1 - \int_0^{3.84} f(x; 1, \lambda(n, \beta^*)) dx$ . Here  $\lambda(n, \beta^*)$  is a function of  $n$  and  $\beta^*$  and is the parameter indexing a non-central  $\chi^2$  distribution with 1 degree of freedom.
- For sample size and MDE calculations, we solve for the design parameter that yields a non-centrality parameter indexing the non-central  $\chi^2$  for which one minus its cumulative distribution function to 3.84 has probability  $\pi$ .
  - When the required power is  $\pi = 80\%$  at 5% significance level, we obtain  $n$  using  $\lambda = 7.85$ ; when the required power  $\pi = 90\%$  at 5% significance level, we obtain  $n$  using  $\lambda = 10.5$ ;
  - When the required power is  $\pi = 80\%$  at 5% significance level, we obtain MDE ( $\beta^*$ ) using  $\lambda = 7.85$ ; When required power  $\pi = 90\%$  at 5% significant level, we obtain MDE ( $\beta^*$ ) using  $\lambda = 10.5$ .

# Power Analysis based upon the Wald test

- Wald Test:

- the non-centrality parameter  $\lambda(\pi) = n \times \frac{(\beta^*)^2}{I_j^{-1}(\hat{\beta}^*)}$  with  $I_j^{-1}(\hat{\beta}^*)$  is the  $r - r$ th element of the covariance matrix  $Var(\hat{\beta})$  of  $\hat{\beta}$  based on the expected information matrix for a single observation  $I_j(\hat{\beta})$ .
- Sample size  $n$  can be found as the solution to  $n = \lambda(\pi) \times \frac{I_j^{-1}(\hat{\beta}^*)}{(\beta^*)^2}$
- MDE  $\beta^*$  can be found as the solution to using  $\lambda(\pi) = n \times \frac{(\beta^*)^2}{I_j^{-1}(\hat{\beta}^*)}$  given  $n$ ,  $\pi$ ,  $\alpha$  and  $\beta_0$ .

# Hypothesis testing in the presence of nuisance parameters

- Nearly always in likelihood-based hypothesis testing, there is one or more parameter(s) of interest, and one or more nuisance parameters.
  - For example, in a logistic regression model for a individually randomized trial,

$$\text{logit}\left(P(Y = 1|X)\right) = \beta_0 + \beta^* \times \mathbf{I}(k = 2)$$

the parameter of interest is the slope,  $\beta^*$ , and the nuisance parameter is the intercept,  $\beta_0$

- In a linear regression model for an individually randomized trial,

$$Y = \beta_0 + \beta^* \times \mathbf{I}(k = 2) + \epsilon, \quad \epsilon \sim N(0, \sigma^2),$$

there are typically two nuisance parameters, the intercept  $\beta_0$  and the variance  $\sigma^2$ .

- All 3 asymptotic tests statistics for  $H_0 : \beta^* = 0$  are a function of these nuisance parameters.
- For testing after a study has been conducted, estimates of the nuisance parameters are substituted in where needed in the test statistic.

# Hypothesis testing in the presence of nuisance parameters

- For power, sample size and MDE calculations, which take place before the study is conducted, values of the <https://www.overleaf.com/project/610fff75652c3e5fd6580767> nuisance parameters must be obtained in other ways, often from pilot data or from the literature. This is typically done for variance component and correlation parameters.
- For power calculations based on the likelihood-based tests in multi-parameter models, the null variance, that is, the variance of  $\hat{\beta}^*$  when under the null when  $\beta^* = 0$  can be used. Then, it is necessary to solve for the convergent value of  $\hat{\beta}_0$  under the null and this is plugged into the variance matrix wherever it appears. In randomized studies, this would be the baseline mean, or the mean of the outcome in the control group, or a function thereof, depending upon the link function.

# Hypothesis testing in the presence of nuisance parameters

- Alternatively, the variance under the alternative be used. Then, in a randomized trial with randomization probability  $p_x$ ,  $1 - p_x$  of the information components use the null variance and  $p_x$  of the information components use the alternative variance. Then, it's necessary to solve for the value of the intercept,  $\beta_0$ , under the specified alternative value for  $\beta^*$ .
  - For example, for continuous data, given the baseline outcome mean or mean of the outcome in the control group,  
$$E(Y|X = 0) = \mu_1 = \beta_0 + \beta^* \implies \beta_0 = \mu_1 - \beta^*.$$
  - For example, for logistic regression with baseline outcome rate, or outcome rate in the control group,  $p_1$ , we solve for  $\beta_0$  as  
$$p_1 = \text{expit}\{\beta_0 + \beta^*\} \implies \beta_0 = .$$

# Power Analysis based upon the Wald test

- In the logistic regression example:

- $X \sim \text{Bern}(p_x)$  and  $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$

- 

$$l_j(\hat{\beta}) = E_X \left[ E_{Y|X} \left\{ - \frac{\partial^2 \log f(Y_j | X_j; \beta)}{\partial \beta \partial \beta^T} \bigg|_{\beta = \hat{\beta}} \right\} \right]$$
$$= \begin{bmatrix} l_{j, \beta_0} & l_{j, \beta_0 \beta^*} \\ l_{j, \beta_0 \beta^*} & l_{j, \beta^*} \end{bmatrix}$$

where

$$l_{j, \beta_0} = E_X \left[ \frac{\exp(\hat{\beta}_0 + \hat{\beta}^* X_j)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^* X_j)\}^2} \right] =$$

$$p_x \frac{\exp(\hat{\beta}_0 + \hat{\beta}^*)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^*)\}^2} + (1 - p_x) \frac{\exp(\hat{\beta}_0)}{\{1 + \exp(\hat{\beta}_0)\}^2},$$

$$l_{j, \beta^*} = E_X \left[ \frac{\exp(\hat{\beta}_0 + \hat{\beta}^* X_j)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^* X_j)\}^2} X_j^2 \right] = p_x \frac{\exp(\hat{\beta}_0 + \hat{\beta}^*)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^*)\}^2}$$

$$l_{j, \beta_0 \beta^*} = E_X \left[ \frac{\exp(\hat{\beta}_0 + \hat{\beta}^* X_j)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^* X_j)\}^2} X_j \right] = p_x \frac{\exp(\hat{\beta}_0 + \hat{\beta}^*)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^*)\}^2}.$$

# Power Analysis based upon the Wald test

- $Var(\hat{\beta}) = I_j^{-1}(\hat{\beta})$ , which is a 2 by 2 matrix inversion

- The 2-2th element in  $I_j^{-1}(\hat{\beta}_{H_A})$  is :

$$I_j^{-1}(\hat{\beta}_{H_A}) = \frac{1}{1-p_x} \frac{\{1+\exp(\beta_{0|H_0})\}^2}{\exp(\beta_{0|H_0})} + \frac{1}{p_x} \frac{\{1+\exp(\beta_{0|H_A}+\beta^*)\}^2}{\exp(\beta_{0|H_A}+\beta^*)},$$

where  $\beta_{0|H_A}$  is solved for as described above

# Power Analysis based upon the Score test

- Recall that the score test is computed as the quadratic form  $\mathbf{U}^T \mathbf{I}^{-1} \mathbf{U}$  using the observed data but fixing the parameters to be those assumed under  $H_0$
- For power analysis, we set  $\beta_{H_0}$  is the  $\beta_0$  and  $\beta^*$  under  $H_0$ , where often  $\beta^* = 0$ , and  $\beta_{H_A} = (\beta_{0|H_A}, \beta^*)^T$  under  $H_A$ . If it is too difficult to solve for  $\beta_{0|H_A}$ , we may set  $\beta_{0|H_A} = \beta_{0|H_0}$ , although this is actually inconsistent with the actual alternative distribution.
- We denote the  $r$ th element in the score function  $\mathbf{U}$  as  $U_{\beta^*}$  for  $\beta^*$ , and the score function for  $\beta_0$  as  $U_{\beta_0}$ . Similarly, if there are additional parameters, e.g. mean and covariance parameters.

# Power Analysis based upon the score test

- To conduct power analysis based upon the score test, the expected value of the non-centrality parameter,  $\lambda(\pi)$  must be derived as a function of  $n$ ,  $\beta^*$  and other design parameters, such as  $E(Y|H_0)$ ,  $\beta_{0|H_A}$  and  $p_x$
- It has been shown that the non-centrality parameter,  $\lambda(\pi) = \xi^2 / I^*(\hat{\beta}^*)$ , where  $\xi$  is the expectation of

$$U_{\beta^*}(\beta_{0|H_A}, \beta_{H_0}^*) - I_{\beta_0\beta^*}^{-1} I_{\beta_0\beta_0}^{-1} U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*) \quad (7)$$

and  $I^*(\hat{\beta}^*)$  is the variance of (7) with respect to  $(Y, X)$  evaluated at  $\tilde{\beta}_0$  and  $\beta_{H_0}^*$ .

- The general formula for  $\lambda(\pi)$  is given by Self (1988)<sup>§</sup>
- As always, sample size  $n$  can be solved using  $\lambda(\pi) = \xi^2 / I^*(\beta^*)$  given  $\lambda(\pi)$  and  $\beta^*$ ,  $\beta_0$  at  $H_0$ ,  $\beta_0$  at  $H_A$ .
- MDE  $\beta^*$  can be solved using  $\lambda(\pi) = \xi^2 / I^*(\hat{\beta}^*)$  given  $n$ ,  $\lambda(\pi)$  and  $\beta_0$  at  $H_0$ ,  $\beta_0$  at  $H_A$ .

<sup>§</sup>Self, Steven G., and Robert H. Mauritsen. "Power/sample size calculations for generalized linear models." Biometrics (1988):

# The score test: Logistic Regression

Now, we need to find

$\lambda(\pi) = \xi^2 / I^*(\hat{\beta}^*) = E_{Y,X}[U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1} U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*)]$  and  $I^*(\hat{\beta}^*) = \text{Var}\{U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1} U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*)\}$  with respect to  $(Y, X)$  evaluated at  $\tilde{\beta}_0$  and  $\beta_{H_0}^* = 0$ .

- First, we find  $\tilde{\beta}_0$ , then  $U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*)$ , then  $I_{\beta_0\beta^*}$  and then  $I_{\beta_0\beta_0}$ . We can put them all together to obtain  $\lambda(\pi)$ .

$\tilde{\beta}_0$ , is the solution of  $0 = \lim_{n \rightarrow \infty} \frac{1}{n} U_{\beta_0}(\beta_{0|H_A}, \beta_{H_0}^*)$ , and its estimator is  $\hat{\beta}_{0|H_0}$ .

- With  $X \sim \text{Bern}(p_x)$ , and  $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$ , based on (1)

- $Pr(Y = 1|X = 0) = p_0 = \frac{\exp(\beta_{0|H_A})}{1 + \exp(\beta_{0|H_A})}$  and

$$Pr(Y = 1|X = 1) = p_1 = \frac{\exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\beta_{0|H_A} + \beta^*)}$$

- $p^* = (1 - p_x)p_0 + p_x p_1$  and  $\tilde{\beta}_0$  is *logit*( $p^*$ ) (Liu and Liang, 1997\*).

<sup>†</sup>Liu, Guanghan, and Kung-Yee Liang. "Sample size calculations for studies with correlated observations." *Biometrics* (1997): 937-947.

# The score test: Logistic Regression

- Next we find  $U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*)$  and  $U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*)$

$$U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) = \sum_{j=1}^n X_i \left\{ Y_i - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \right\} = \sum_{j=1}^n X_i (Y_i - p^*)$$

and

$$U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*) = \sum_{j=1}^n \left\{ Y_i - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \right\} = \sum_{j=1}^n (Y_i - p^*)$$

# The score test: Logistic Regression

- Next we find  $I_{\beta_0\beta^*}$  and  $I_{\beta_0\beta_0}$ :

$$\begin{aligned} I_{\beta_0\beta^*} &= E \left\{ \frac{\partial^2 p_i}{\partial \beta^* \beta_0} \right\} \Big|_{\beta_{H_0}^* \text{ and } \tilde{\beta}_0} = E_X[E_{Y|X} \{X_i p_j (1 - p_j)\}] \Big|_{\beta_{H_0}^* \text{ and } \tilde{\beta}_0} \\ &= p_x \times \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \left\{ 1 - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \right\} \\ &= p_x p^* (1 - p^*) \end{aligned} \quad (8)$$

and

$$\begin{aligned} I_{\beta_0\beta_0} &= E \left\{ \frac{\partial^2 p_i}{\partial \beta_0^2} \right\} \Big|_{\beta_{H_0}^* \text{ and } \tilde{\beta}_0} = E_X[E_{Y|X} \{p_j (1 - p_j)\}] \Big|_{\beta_{H_0}^* \text{ and } \tilde{\beta}_0} \\ &= \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \left\{ 1 - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \right\} \\ &= p^* (1 - p^*) \end{aligned} \quad (9)$$

# The score test: Logistic Regression

- Then, the numerator of the non-centrality parameter,

$$\begin{aligned}\xi &= E_{Y,X}[U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1} U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*)] \\&= E_{X,Y} \sum_{j=1}^n (Y_i - p^*)(X_i - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1}) \\&= n E_X[E_{Y|X}\{(Y_i - p^*)(X_i - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1})\}] \\&= n E_X[(p_i - p^*)(X_i - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1})] \\&= n [p_x\{(p_1 - p^*)(1 - p_x)\} - (1 - p_x)\{(p_0 - p^*)p_x\}] \\&= n * p_x(1 - p_x)(p_1 - p_0)\end{aligned}$$

where  $p^* = (1 - p_x)p_0 + p_x p_1$ ,  $I_{\beta_0\beta_0}$  is given by (8), and  $I_{\beta_0\beta^*}$  is given by (9).

# The score test: Logistic Regression

- The denominator of the non-centrality parameter,

$$\begin{aligned} I_{\beta^*}^* &= nE_X\{p_i(1-p_i)(X_i - I_{\beta_0\beta^*}I_{\beta_0\beta_0}^{-1})^2\} \\ &= n[p_x\{p_1(1-p_1)(1-p_x)^2\} + p_0(1-p_0)(1-p_x)(-p_x)^2] \\ &= np_x(1-p_x)\{(1-p_x)p_1(1-p_1) + p_xp_0(1-p_0)\} \end{aligned}$$

- As a result, we have

$$\begin{aligned} \lambda(\pi) &= \xi^2 / I^*(\beta^*) = \frac{\{n \times p_x(1-p_x)(p_1-p_0)\}^2}{np_x(1-p_x)\{(1-p_x)p_1(1-p_1) + p_xp_0(1-p_0)\}} \\ &= n \frac{(p_1-p_0)^2 p_x(1-p_x)}{(1-p_x)p_1(1-p_1) + p_xp_0(1-p_0)} \end{aligned} \quad (10)$$

- For sample size  $n$ , we have  $n = \frac{\lambda(\pi) \times (1-p_x)p_1(1-p_1) + p_xp_0(1-p_0)}{(p_1-p_0)^2 p_x(1-p_x)}$  with  $\lambda(80\%) = 7.85$  and  $\lambda(90\%) = 10.5$  at 5% significant level.
- For MDE, we solve (10) given  $p_x$ ,  $\beta_{0|H_A}$ ,  $\lambda(\pi)$  and  $n$ .

# Power Analysis based upon the LRT

- Similar to the score test, we need to approximate the distribution of the test statistic

$$Z^2 = 2\{\log f(\mathbf{Y}|\mathbf{X}; \hat{\beta}_{H_A}) - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \hat{\beta}_{-r|H_0})\}$$

under  $H_A$  (Self, 1992<sup>¶</sup>).

- To be specific, we decompose  $Z^2$  into three terms:

$$\begin{aligned} Z^2 = & 2\{\log f(\mathbf{Y}|\mathbf{X}; \hat{\beta}_{H_A}) - \log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})\} \\ & - 2\{\log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \hat{\beta}_{-r|H_0}) - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \tilde{\beta}_{-r})\} \\ & + 2\{\log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A}) - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \tilde{\beta}_{-r})\} \end{aligned} \quad (11)$$

where  $\tilde{\beta}_{-r}$  can be obtained similar to what we described in page 47 for the score test, and its estimator is  $\hat{\beta}_{-r|H_0}$ .

- When  $\beta = (\beta_0, \beta^*)$ ,  $\tilde{\beta}_{-r} = \tilde{\beta}_0$  and  $\hat{\beta}_{0|H_0}$  is the estimator of  $\tilde{\beta}_0$ .
- An approximate expected value for the first term in (11) is the number of parameters in the model. In the case of  $\beta = (\beta_0, \beta^*)$ , it takes value of 2.

<sup>¶</sup>Self, Steven G., Robert H. Mauritsen, and Jill Ohara. "Power calculations for likelihood ratio tests in generalized linear models." *Biometrics* (1992): 31-39.

# Power Analysis based upon the LRT

- An approximation of the expected value of the second term in (11) equals to a **trace** of matrix  $M$ , where

$$M = \left[ E_{\mathbf{Y}|\mathbf{X}} \left\{ \frac{\partial^2 \log f(\mathbf{Y}|\mathbf{X}; \boldsymbol{\beta})}{\partial \boldsymbol{\beta}_{-r} \partial \boldsymbol{\beta}_{-r}^T} \middle|_{(\tilde{\boldsymbol{\beta}}_{-r}, \boldsymbol{\beta}_{H_0}^*)} \right\} \right]^{-1} E_{\mathbf{Y}|\mathbf{X}} \left[ \frac{\partial \log f(\mathbf{Y}|\mathbf{X}; \boldsymbol{\beta})}{\partial \boldsymbol{\beta}_{-r}} \middle|_{(\tilde{\boldsymbol{\beta}}_{-r}, \boldsymbol{\beta}_{H_0}^*)} \right]^{\otimes 2},$$

where for a vector  $\mathbf{a}$ ,  $\mathbf{a}^{\otimes 2} = \mathbf{a}\mathbf{a}^T$ .

- The first term of  $M$  is the expectation of the second derivative of log-likelihood function evaluated at  $(\tilde{\boldsymbol{\beta}}_{-r}, \boldsymbol{\beta}_{H_0}^*)$ .
- The second term of  $M$  is the variance of the first derivative of log-likelihood function evaluated at  $(\tilde{\boldsymbol{\beta}}_{-r}, \boldsymbol{\beta}_{H_0}^*)$ .
- Under the setting with  $\boldsymbol{\beta} = (\beta_0, \beta^*)^T$ ,  $M$  is a scalar and  $\text{trace}(M) = M$ .

# Power Analysis based upon the LRT

- The expected value of the third term in (11), denoted by  $\Delta$ , can be calculated exactly:

$$\Delta = E_{\mathbf{Y}|\mathbf{X}}[2\{\log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})\} - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \tilde{\beta}_{-r})\}]$$

- The expected value of a noncentral  $\chi^2$  random variable with 1 degree of freedom and noncentrality parameter  $\lambda(\pi)$  is  $1 + \lambda(\pi)$ .
- Equating this to the approximation derived from (11), we can obtain

$$\lambda(\pi) = \Delta + 1 - \text{trace}(M)$$

- Under the setting with  $\beta = (\beta_0, \beta^*)^T$ ,

$$\lambda(\pi) = \Delta + 1 - M$$

# LRT: Logistic Regression Example

- $X \sim \text{Bern}(p_x)$  and  $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$
- $Pr(Y = 1|X = 0) = p_0 = \frac{\exp(\beta_{0|H_A})}{1 + \exp(\beta_{0|H_A})}$  and  
 $Pr(Y = 1|X = 1) = p_1 = \frac{\exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\beta_{0|H_A} + \beta^*)}$
- $p^* = (1 - p_x)p_0 + p_x p_1$  and  $\tilde{\beta}_0$  is *logit*( $p^*$ )
- To find  $\lambda(\pi)$ , we need to calculate  $\Delta$  and  $(M)$ .

1. We first calculate

$\Delta = E_{\mathbf{YX}}[2\{\log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})\} - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \tilde{\beta}_0)]$  by using

$$E_{\mathbf{YX}}[\log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})] = E_{\mathbf{X}}[E_{\mathbf{Y}|\mathbf{X}}[\log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})]]$$

$$= nE_{\mathbf{X}}\left[\frac{\exp(\beta_{0|H_A} + \beta^* X_j)}{1 + \exp(\beta_{0|H_A} + \beta^* X_j)}(\beta_{0|H_A} + \beta^* X_j) - \log\{1 + \exp(\beta_{0|H_A} + \beta^* X_j)\}\right]$$

$$= np_x \left[ \frac{\exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\beta_{0|H_A} + \beta^*)}(\beta_{0|H_A} + \beta^*) - \log\{1 + \exp(\beta_{0|H_A} + \beta^*)\} \right]$$

$$+ n(1 - p_x) \left[ \frac{\exp(\beta_{0|H_A})}{1 + \exp(\beta_{0|H_A})}(\beta_{0|H_A}) - \log\{1 + \exp(\beta_{0|H_A})\} \right]$$

# LRT: Logistic Regression Example

$$\begin{aligned} E_{\mathbf{Y}|\mathbf{X}}[\log f(\mathbf{Y}|\mathbf{X}; \tilde{\beta}_0, \beta^* = 0)] &= E_{\mathbf{X}}[E_{\mathbf{Y}|\mathbf{X}}[\log f(\mathbf{Y}|\mathbf{X}; \tilde{\beta}_0, \beta^* = 0)]] \\ &= nE_{\mathbf{X}}\left[\frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}(\tilde{\beta}_0) - \log\{1 + \exp(\tilde{\beta}_0)\}\right] \\ &= np_x\left[\frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}(\tilde{\beta}_0) - \log\{1 + \exp(\tilde{\beta}_0)\}\right] \\ &\quad + n(1 - p_x)\left[\frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}(\tilde{\beta}_0) - \log\{1 + \exp(\tilde{\beta}_0)\}\right] \end{aligned}$$

Then we have

$$\begin{aligned} \Delta &= 2n\left[p_x\left\{\frac{\exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\beta_{0|H_A} + \beta^*)}(\beta_{0|H_A} + \beta^*) - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}\tilde{\beta}_0\right.\right. \\ &\quad \left.- \log\left(\frac{1 + \exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\tilde{\beta}_0)}\right)\right\} + (1 - p_x)\left\{\frac{\exp(\beta_{0|H_A})}{1 + \exp(\beta_{0|H_A})}\beta_{0|H_A}\right. \\ &\quad \left.- \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}\tilde{\beta}_0 - \log\left(\frac{1 + \exp(\beta_{0|H_A})}{1 + \exp(\tilde{\beta}_0)}\right)\right\}\right] \end{aligned}$$

# LRT: Logistic Regression Example

$\Delta$  can be further simplified as

$$\Delta = 2n \left[ p_x \left\{ p_1(\beta_{0|H_A} + \beta^* - \tilde{\beta}_0) - \log \left( \frac{1 + \exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right. \\ \left. + (1 - p_x) \left\{ p_0(\beta_{0|H_A} - \tilde{\beta}_0) - \log \left( \frac{1 + \exp(\beta_{0|H_A})}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right]$$

We let  $\Delta = n\Delta^*$  with

$$\Delta^* = 2 \left[ p_x \left\{ p_1(\beta_{0|H_A} + \beta^* - \tilde{\beta}_0) - \log \left( \frac{1 + \exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right. \\ \left. + (1 - p_x) \left\{ p_0(\beta_{0|H_A} - \tilde{\beta}_0) - \log \left( \frac{1 + \exp(\beta_{0|H_A})}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right]$$

# LRT: Logistic Regression Example

2. Next, we can obtain

$$M = E_{\mathbf{X}\mathbf{Y}} \left[ - \frac{\partial^2 \log f(\mathbf{Y}|\mathbf{X}; \beta)}{\partial \beta_0^2} \bigg|_{(\tilde{\beta}_0, \beta_{H_0}^*)} \right]^{-1} E_{\mathbf{X}\mathbf{Y}} \left[ \left\{ \frac{\partial \log f(\mathbf{Y}|\mathbf{X}; \beta)}{\partial \beta_0} \bigg|_{(\tilde{\beta}_0, \beta_{H_0}^*)} \right\}^2 \right]:$$

• The second term in  $M$  is

$$\begin{aligned} E_{\mathbf{X}\mathbf{Y}} \left[ \left\{ \frac{\partial \log f(\mathbf{Y}|\mathbf{X}; \beta)}{\partial \beta_0} \bigg|_{(\tilde{\beta}_0, \beta_{H_0}^*)} \right\}^2 \right] &= E_{\mathbf{X}\mathbf{Y}} \left[ \left\{ \sum_{j=1}^n Y_j - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \right\}^2 \right] \\ &= E_{\mathbf{X}\mathbf{Y}} \left[ \left\{ \sum_{j=1}^n Y_j - np^* \right\}^2 \right] \\ &= E_{\mathbf{X}\mathbf{Y}} \left[ \left( \sum_{j=1}^n Y_j \right)^2 \right] - 2np^* E_{\mathbf{X}\mathbf{Y}} \left[ \left( \sum_{j=1}^n Y_j \right) \right] + n(p^*)^2 \\ &= n\text{Var}(Y_j) - 2n^2 p^* E_{\mathbf{X}}(p_j) + n(p^*)^2 \\ &= n\{p_x p_1(1 - p_1) + (1 - p_x)p_0(1 - p_0)\} - 2n^2 p^* \{p_x p_1 + (1 - p_x)p_0 - p^*\} \\ &= n\{p_x p_1(1 - p_1) + (1 - p_x)p_0(1 - p_0)\} \end{aligned}$$

# LRT: Logistic Regression Example

- The first term in  $M$  is  $E_{\mathbf{X}\mathbf{Y}} \left[ - \frac{\partial^2 \log f(\mathbf{Y}|\mathbf{X};\boldsymbol{\beta})}{\partial \beta_0^2} \Big|_{(\tilde{\beta}_0, \beta_{H_0}^*)} \right]^{-1} = \left\{ n \frac{\exp(\tilde{\beta}_0)}{\{1 + \exp(\tilde{\beta}_0)\}^2} \right\}^{-1}$
- Thus

$$M = \frac{p_x p_1 (1 - p_1) + (1 - p_x) p_0 (1 - p_0)}{p^* (1 - p^*)}$$

$$\text{with } p_1 = \frac{\exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\beta_{0|H_A} + \beta^*)}, p_0 = \frac{\exp(\beta_{0|H_A})}{1 + \exp(\beta_{0|H_A})} \text{ and } p^* = \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}.$$

# LRT: Logistic Regression Example

- As a result,

$$\begin{aligned}\lambda(\pi) &= n\Delta^* + 1 - M \\ &= 2n \left[ p_x \left\{ p_1(\beta_{0|H_A} + \beta^* - \tilde{\beta}_0) - \log \left( \frac{1 + \exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right. \\ &\quad \left. + (1 - p_x) \left\{ p_0(\beta_{0|H_A} - \tilde{\beta}_0) - \log \left( \frac{1 + \exp(\beta_{0|H_A})}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right] \\ &\quad + 1 - \frac{p_x p_1(1 - p_1) + (1 - p_x)p_0(1 - p_0)}{p^*(1 - p^*)}\end{aligned}\tag{12}$$

- For sample size  $n$ , we have  $n = \frac{\lambda(\pi) - 1 + M}{\Delta^*}$  with  $\lambda(80\%) = 7.85$  and  $\lambda(90\%) = 10.5$  at 5% significant level.
- For MDE, we solve (12) given  $p_x$ ,  $\beta_{0|H_A}$ ,  $\tilde{\beta}_0$ ,  $\lambda(\pi)$  and  $n$ .

## Example: HPTN 037

In this example, we will use the value and estimates obtained from HPTN 037 study to calculate the following items based on three hypothesis tests.

- the power of HPTN to detect an overall intervention effect corresponding to an RR of 0.5 and 0.75, given the baseline rate and sample size
- Then calculate sample size to detect an RR of 0.5 and 0.75 with 80% and 90% power
- Then calculate minimum detectable RR and risk difference (RD) given the sample size and power

## Example: HPTN 037 Choosing design parameters

Recall that  $n = 696$  in HPTN 037. We will use estimated values from the data to guide our hypothetical exploration of study design option.

$p_x = \hat{P}(X = 1) = 336/696 = 0.483$ , intended to be 0.5 by design.

Under  $H_A$ ,  $\hat{\beta} = (-0.022, -0.314)$ .

Under  $H_0$ ,  $\hat{\beta}_0 = -0.173$ .

These  $\hat{\beta}$  give us an estimated  $\hat{Pr}(Y = 1|X = 0) = \hat{p}_0 = \frac{\exp\{-0.022\}}{1 + \exp\{-0.022\}} = 0.495$ , the baseline outcome rate, and  $\hat{Pr}(Y = 1) = \hat{p}^* = \frac{\exp\{-0.173\}}{1 + \exp\{-0.173\}} = 0.457$ . This latter is the post-intervention average over  $X=0$  and  $X=1$ .

$$\hat{RR} = \exp(\hat{\beta}^*) = \exp\{-0.314\} = 0.731$$

This is how we obtain approximate values for study design from existing data

# Power Calculation: Wald Test

For  $RR = \exp(\beta^*) = 0.5$

- We let  $n = 696$ ,  $p_x = 0.483$ ,  $\beta_0 = -0.022$ , and  $\beta^* = \log(0.5)$ , then  $\lambda = n * \frac{(\beta^*)^2}{I_j^{-1}(\beta^*)} = 19.53$ , the formula is given in page 44.
- Power =  $\Pr(X_1^2(\lambda = 19.53) > 3.84) = 99.3\%$

```
> p_x = 0.483; b0 = -0.022 ; b1=log(0.5); n=696
> # I^{-1}(\beta^*):
> Ib1 = (1/(1-p_x))*((1+exp(b0))^2/(exp(b0)))+
+       (1/p_x)*((1+exp(b0+b1))^2/(exp(b0+b1)))
> Ib1
[1] 17.12423
> lambda_w = n*(b1^2)/Ib1
> lambda_w
[1] 19.52761
> power = 1-pchisq(3.84, 1, ncp = lambda_w, lower.tail = TRUE,)
> power
[1] 0.9930418
```

## Numeric Results for Two-Sided Wald Test

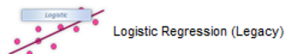
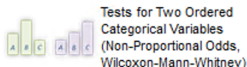
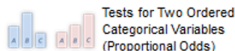
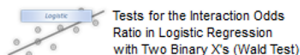
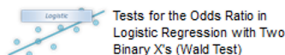
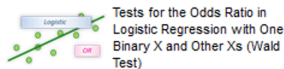
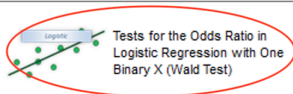
Alternative Hypothesis:  $OR_{yx} \neq 1$

| Power   | N   | Percent |  | P0     | P1      | OR <sub>yx</sub> | Alpha | Beta    |
|---------|-----|---------|--|--------|---------|------------------|-------|---------|
|         |     | X=1     |  |        |         |                  |       |         |
| 0.99303 | 696 | 48.3    |  | 0.4945 | 0.32846 | 0.5              | 0.05  | 0.00697 |

# Wald Test-PASS results

- Using  $RR = 0.5$  as an example, to obtain the PASS results, we can follow steps:
  - Step 1: Find the module: logistic regression with one binary X(Wald test)

## Regression > Logistic Regression > Binary X (Wald Test) (7)



# Wald Test-PASS results

- Step 2: We select it solve for Power and input  $n$ ,  $\alpha$ ,  $p_0$ ,  $RR$  and  $p_x$

Tests for the Odds Ratio in Logistic Regression with One Binary X (Wald Test) (Licensed for Student Use Only)

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**Calculate**

**Design**

Solve For: Power

Y = Disease 1 = Yes  
X = Exposure 0 = No

**Test**

Alternative Hypothesis: Two-Sided

**Alpha**

Alpha: 0.05

**Sample Size**

N (Sample Size): 696

**Baseline Probability**

P0 [Pr(Y = 1 | X = 0)]: 0.4945

**P1 or Odds Ratio**

Use P1 or ORyx: ORyx

ORyx (Y,X Odds Ratio): 0.5

**Prevalence of X**

Percent with X = 1: 48.3

**Help Center**

For this procedure:

- Documentation
- Examples
- Validation Examples
- Open Example Template
- Validate This Procedure

**General:**

- Getting Started Video
- Creating Power Curves Video
- All Training Videos

**Option Info**

**ORyx (Y,X Odds Ratio)**

Specify one or more values of the Odds Ratio of Y and X, a measure of the effect size (event rate) that is to be detected by the study.

**Odds Ratio of Y and X**

This is the ratio of the odds of the outcome Y given that the exposure X = 1 to the odds of Y = 1 given X = 0.

$$\text{Odds}(Y=1|X=1) / \text{Odds}(Y=1|X=0)$$

**Odds(Event A)**

$$\text{Odds}(A) = \text{Prob}(A) / \text{Prob}(\text{Not } A)$$

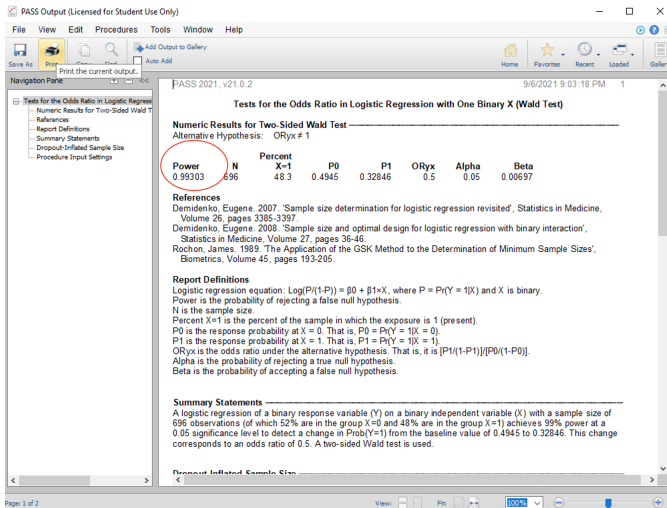
**Input**

You can enter a single value such as

1.5

# Wald Test-PASS results

- Step 3: Click "Calculate", we will have the results



# Power Calculation: Wald Test

For  $RR = \exp(\beta^*) = 0.75$

- We let  $n = 696$ ,  $p_x = 0.483$ ,  $\beta_0 = -0.022$ , and  $\beta^* = \log(0.75)$ , then  $\lambda = n * \frac{(\beta^*)^2}{I_j^{-1}(\beta^*)} = 3.551$ ,
- $\text{power} = Pr(X_1^2(\lambda = 3.551) > 3.84) = 47.01\%$

```
> p_x = 0.483; b0 = -0.022 ; b1=log(0.75); n=696
> # I^{-1}(\beta^*):
> Ib1 = (1/(1-p_x))*((1+exp(b0))^2/(exp(b0))) +
+ (1/p_x)*((1+exp(b0+b1))^2/(exp(b0+b1)))
> Ib1
[1] 16.2196
> lambda_w = n*(b1^2)/Ib1
> lambda_w
[1] 3.551359
> power = 1-pchisq(3.84, 1, ncp = lambda_w, lower.tail = TRUE,)
> power
[1] 0.4701334
```

## Numeric Results for Two-Sided Wald Test

Alternative Hypothesis:  $OR_{yx} \neq 1$

| Power   | N   | Percent<br>X=1 | P0     | P1      | OR <sub>yx</sub> | Alpha | Beta    |
|---------|-----|----------------|--------|---------|------------------|-------|---------|
| 0.46999 | 696 | 48.3           | 0.4945 | 0.42319 | 0.75             | 0.05  | 0.53001 |

# Power Calculation for the Score Test

For  $RR = \exp(\beta^*) = 0.5$ , then  $\beta^* = \log(0.5)$

- We let  $n = 696$ ,  $p_x = 0.483$ ,  $\beta_{0|H_0} = -0.173$ ,  $\beta_{0|H_A} = -0.022$ , then  
$$\lambda = \xi^2 / I^*(\beta^*) = n \times \frac{p_x(1-p_x)(p_1-p_0)^2}{p_x p_0(1-p_0) + (1-p_x)p_1(1-p_1)} = 20.41,$$
- power =  $Pr(X_1^2(\lambda = 20.41) > 3.84) = 99.5\%$

```
> p_x = 0.483; b00 = -0.173; b0a = -0.022; b1=log(0.5) ; n=696#
> p_0 =exp(b00)/(1+exp(b00)); # p_y0 =exp(b00)/(1+exp(b00));
> p_1 = exp(b0a+b1)/(1+exp(b0a+b1))
> lambda_s = n*(p_x*(1-p_x)*(p_1-p_0)^2)/(p_x*p_0*(1-p_0)+ (1-p_x)*p_1*(1-p_1))
> lambda_s
[1] 20.40859
> power = 1-pchisq(3.84, 1, ncp = lambda_s, lower.tail = TRUE, log.p = FALSE)
> power
[1] 0.9947361
> |
```

For  $RR = \exp(\beta^*) = 0.75$ , then  $\beta^* = \log(0.75)$

- We let  $n = 696$ ,  $p_x = 0.483$ ,  $\beta_{0|H_0} = -0.173$ ,  $\beta_{0|H_A} = -0.022$ , then  
$$\lambda = \xi^2 / I^*(\beta^*) = n \times \frac{p_x(1-p_x)(p_1-p_0)^2}{p_x p_0(1-p_0) + (1-p_x)p_1(1-p_1)} = 3.579,$$
- power =  $Pr(X_1^2(\lambda = 3.579) > 3.84) = 47.3\%$

# Power Calculation: LRT

For  $RR = \exp(\beta^*) = 0.5$

- We let  $n = 696$ ,  $p_x = 0.483$ ,  $\beta_{0|H_A} = -0.022$ ,  $\tilde{\beta}_0 = -0.346$  and  $\beta^* = \log(0.5)$ ,  $\Delta^* = 0.0286$  and  $M = 0.972$  with formula of  $\Delta^*$  in page 57 and formula of  $M$  in page 59.
- $\lambda = 19.908$  using formula in page 60, and power = 99.38%

```
> p_x = 0.483; b0a = -0.022; b1=log(0.5); n=696
> p0 =exp(b0a)/(1+exp(b0a)); p1 = exp(b0a+b1)/(1+exp(b0a+b1))
> p_star = (1-p_x)*p0+p_x*p1; b_tilde = log(p_star/(1-p_star))
> b_tilde
[1] -0.3462009
> D_star = 2*( p_x*(p1*(b0a+b1)-p_star*b_tilde -
+ log((1+exp(b0a+b1))/(1+exp(b_tilde))) ) )
+ (1-p_x)* (p0*b0a-p_star*b_tilde-
+ log((1+exp(b0a))/(1+exp(b_tilde))) ) )
> D_star
[1] 0.0285624
> M_t = (p_x*(1-p1)*p1 +(1-p_x)*p0*(1-p0))/
+ (p_star *(1-p_star) )
> M_t
[1] 0.97163
> lambda = n*D_star+1-M_t
> lambda
[1] 19.9078
> power=1-pchisq(3.84, 1, ncp = lambda,lower.tail = TRUE)
> power
[1] 0.9938292
```

For  $RR = \exp(\beta^*) = 0.75$

- We let  $n = 696$ ,  $p_x = 0.483$ ,  $\beta_{0|H_A} = -0.022$ ,  $\tilde{\beta}_0 = -0.160$  and  $\beta^* = \log(0.75)$ ,  $\Delta^* = 0.005$  and  $M = 0.995$
- $\lambda = 3.567$  and power = 47.18%

# Sample Size Calculation: For Wald Test

When  $RR = \exp(\beta^*) = 0.50$

- We let power = 80%,  $p_x = 0.483$ ,  $\beta_0 = -0.022$ , and  $\beta^* = \log(0.5)$ , then  $n = 280$  using formula in page 44.
- We let power = 90%, then  $n = 375$ .

```
p_x = 0.483;b0=-0.022;b1 = log(0.5)
Ib1 = (1/(1-p_x))*((1+exp(b0))^2/(exp(b0)))+
      (1/p_x)*((1+exp(b0+b1))^2/(exp(b0+b1)))
n_80=7.85/((b1^2)/Ib1)
n_90 = 10.5/((b1^2)/Ib1)
n_80
[1] 279.7885
n_90
[1] 374.2394
```

## Numeric Results for Two-Sided Wald Test

Alternative Hypothesis: ORyx ≠ 1

| Power   | N   | Percent<br>X=1 | P0     | P1      | ORyx | Alpha | Beta    |
|---------|-----|----------------|--------|---------|------|-------|---------|
| 0.80035 | 280 | 48.3           | 0.4945 | 0.32846 | 0.5  | 0.05  | 0.19965 |

| Power   | N   | Percent<br>X=1 | P0     | P1      | ORyx | Alpha | Beta    |
|---------|-----|----------------|--------|---------|------|-------|---------|
| 0.90038 | 375 | 48.3           | 0.4945 | 0.32846 | 0.5  | 0.05  | 0.09962 |

# Wald Test - PASS Results

- We use same module: logistic regression with one binary X (Wald test), but in Step 2, we select to solve sample size. Here we input power,  $\alpha$ ,  $p_0$ ,  $RR$  and  $p_x$

Tests for the Odds Ratio in Logistic Regression with One Binary X (Wald Test) (Licensed for Student Use Only)

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**Calculate**

**Design**

Solve For: Sample Size

Y = Disease 1 = Yes  
X = Exposure 0 = No

**Test**

Alternative Hypothesis: Two-Sided

**Power and Alpha**

Power: 0.8

Alpha: 0.05

**Baseline Probability**

P0 [Pr(Y = 1 | X = 0)]: 0.4945

**P1 or Odds Ratio**

Use P1 or ORyx: ORyx

ORyx (Y,X Odds Ratio): 0.5

**Prevalence of X**

Percent with X = 1: 48.3

**Help Center**

For this procedure:

- Documentation
- Examples
- Validation Examples
- Open Example Template
- Validate This Procedure

**General:**

- Getting Started Video
- Creating Power Curves Video
- All Training Videos

**Option Info**

**Power**

Power is the probability of rejecting the null hypothesis when it is false. Power is equal to 1 - Beta, so specifying power implicitly specifies beta.

Beta is the probability of obtaining a false negative with the statistical test. That is, it is the probability of accepting a false null hypothesis.

**Range**

The valid range is 0 to 1. Usually, a power between 0.8 and 0.95 is desired.

**Recommended**

Different disciplines have different standards for setting power. The most common choice is 0.90, but 0.80 is also popular.

# Wald Test - PASS Results

- After click "Calculate", we can obtain the results:

The screenshot displays the PASS 2021, v21.0.2 interface. The main window shows the 'Tests for the Odds Ratio in Logistic Regression with One Binary X (Wald Test)' results. A red circle highlights the 'Power' value of 0.80036 in the 'Numeric Results for Two-Sided Wald Test' table. The table includes columns for Power, N, Percent X=1, P0, P1, ORyx, Alpha, and Beta. Below the table, the 'References' section lists three sources: Demidenko (2007), Demidenko (2008), and Rochon (1989). The 'Report Definitions' section provides mathematical definitions for the variables used in the test. The 'Summary Statements' section concludes that a logistic regression model with a sample size of 280 observations achieves 80% power at a 0.05 significance level to detect a change in the probability of Y=1 from 0.4945 to 0.32846.

**Tests for the Odds Ratio in Logistic Regression with One Binary X (Wald Test)**

**Numeric Results for Two-Sided Wald Test**

Alternative Hypothesis:  $OR_{yx} \neq 1$

| Power   | N   | Percent X=1 | P0     | P1      | ORyx | Alpha | Beta    |
|---------|-----|-------------|--------|---------|------|-------|---------|
| 0.80036 | 280 | 48.3        | 0.4945 | 0.32846 | 0.5  | 0.05  | 0.19965 |

**References**

Demidenko, Eugene. 2007. 'Sample size determination for logistic regression revisited', *Statistics in Medicine*, Volume 26, pages 3385-3397.

Demidenko, Eugene. 2008. 'Sample size and optimal design for logistic regression with binary interaction', *Statistics in Medicine*, Volume 27, pages 36-46.

Rochon, James. 1989. 'The Application of the GSK Method to the Determination of Minimum Sample Sizes', *Biometrics*, Volume 45, pages 193-205.

**Report Definitions**

Logistic regression equation:  $\text{Log}[P/(1-P)] = \beta_0 + \beta_1 \times X$ , where  $P = \text{Pr}(Y = 1|X)$  and  $X$  is binary.

Power is the probability of rejecting a false null hypothesis.

$N$  is the sample size.

Percent  $X=1$  is the percent of the sample in which the exposure is 1 (present).

$P_0$  is the response probability at  $X = 0$ . That is,  $P_0 = \text{Pr}(Y = 1|X = 0)$ .

$P_1$  is the response probability at  $X = 1$ . That is,  $P_1 = \text{Pr}(Y = 1|X = 1)$ .

$OR_{yx}$  is the odds ratio under the alternative hypothesis. That is, it is  $[P_1/(1-P_1)]/[P_0/(1-P_0)]$ .

Alpha is the probability of rejecting a true null hypothesis.

Beta is the probability of accepting a false null hypothesis.

**Summary Statements**

A logistic regression of a binary response variable ( $Y$ ) on a binary independent variable ( $X$ ) with a sample size of 280 observations (of which 52% are in the group  $X=0$  and 48% are in the group  $X=1$ ) achieves 80% power at a 0.05 significance level to detect a change in  $\text{Prob}(Y=1)$  from the baseline value of 0.4945 to 0.32846. This change corresponds to an odds ratio of 0.5. A two-sided Wald test is used.

# For Wald Test

When  $RR = \exp(\beta^*) = 0.75$ ,

- We let power = 80%,  $p_x = 0.483$ ,  $\beta_0 = -0.022$ , and  $\beta^* = \log(0.75)$ , then  $n = 1539$
- We let power = 90%, then  $n = 2058$

```
> p_x = 0.483;b0 = -0.022;b1 = log(0.75)
> Ib1 = (1/(1-p_x))*((1+exp(b0))^2/(exp(b0)))+
+       (1/p_x)*((1+exp(b0+b1))^2/(exp(b0+b1)))
> n_80=7.85/((b1^2)/Ib1)
> n_90 = 10.5/((b1^2)/Ib1)
> n_80
[1] 1538.453
> n_90
[1] 2057.804
```

| Power  | N    | Percent<br>X=1 | P0     | P1      | ORyx | Alpha | Beta   |
|--------|------|----------------|--------|---------|------|-------|--------|
| 0.8002 | 1539 | 48.3           | 0.4945 | 0.42319 | 0.75 | 0.05  | 0.1998 |
| Power  | N    | Percent<br>X=1 | P0     | P1      | ORyx | Alpha | Beta   |
| 0.9001 | 2060 | 48.3           | 0.4945 | 0.42319 | 0.75 | 0.05  | 0.0999 |

# Sample Size Calculation: For Score Test

When  $RR = \exp(\beta^*) = 0.50$

- We let power = 80%,  $p_x = 0.483$ ,  $\beta_{0|H_0} = -0.173$ ,  $\beta_{0|H_A} = -0.022$ , and  $\beta^* = \log(0.5)$ , then  $n = 268$  using formula in page 51 given  $p_1(p_{-y1}$  in the R code) and  $p_0(p_{-y0}$  in the R code).

```
> p_x = 0.483;b00 = -0.173;b0a = -0.022;b1=log(0.5) #
> p_y0 = exp(b0a)/(1+exp(b0a))
> p_y1 = exp(b0a+b1)/(1+exp(b0a+b1))
> n_80=7.85/((p_x*(1-p_x)*(p_y1-p_y0)^2)/
+           (p_x*p_y0*(1-p_y0)+ (1-p_x)*p_y1*(1-p_y1)))
>
> n_90= 10.5/((p_x*(1-p_x)*(p_y1-p_y0)^2)/
+           (p_x*p_y0*(1-p_y0)+ (1-p_x)*p_y1*(1-p_y1)))
> n_80
[1] 267.7108
> n_90
[1] 358.0845
```

- We let power = 90%, then  $n = 359$

When  $RR = \exp(\beta^*) = 0.75$ ,

- We let power = 80%,  $p_x = 0.483$ ,  $\beta_{0|H_0} = -0.173$ ,  $\beta_{0|H_A} = -0.022$ ,  $\beta^* = \log(0.75)$ , then  $n = 1527$
- We let power = 90%, then  $n = 2043$

# Sample Size Calculation: For LRT

When  $RR = \exp(\beta^*) = 0.50$

- We let power = 80%,  $p_x = 0.483$ ,  $\beta_{0|H_A} = -0.022$ ,  $\tilde{\beta}_0 = -0.3462$  and  $\beta^* = \log(0.5)$ , then  $n = 274$ . We let power = 90%, then  $n = 367$  using the formula of  $n$  in page 60 with the formula of  $\Delta^*$  in page 57 and the formula of  $M$  in page 59.

```
> p_x = 0.483; b0a = -0.022; b1=log(0.5); n=696
> p0 =exp(b0a)/(1+exp(b0a)); p1 = exp(b0a+b1)/(1+exp(b0a+b1))
> p_star = (1-p_x)*p0+p_x*p1; b_tilde = log(p_star / (1-p_star ))
> D_star = 2*( p_x*(p1*(b0a+b1)-p_star*b_tilde - log( (1+exp(b0a+b1))/(1+exp(b_tilde)) ) )
+ (1-p_x)* (p0*b0a-p_star*b_tilde- log((1+exp(b0a))/(1+exp(b_tilde)) ) ) )
> D_star
[1] 0.0285624
> M_t = (p_x*(1-p1)*p1 +(1-p_x)*p0*(1-p0))/(p_star *(1-p_star ) )
> M_t
[1] 0.97163
> #power = 80%
> n_80=(7.85-1+M_t)/D_star
> n_80
[1] 273.8436
> #power = 90%
> n_90=(10.5-1+M_t)/D_star
> n_90
[1] 366.6229
```

When  $RR = \exp(\beta^*) = 0.75$ ,

- We let power = 80%, then  $n = 1533$ . We let power = 90%, then  $n = 2051$ .

# MDE Calculation: Wald Test

- We let power = 80%,  $n = 696$ ,  $p_x = 0.483$ ,  $\beta_0 = -0.022$ , then MDE  $\beta^* = -0.431$  then  $RR = \exp(\beta^*) = 0.650$

```
p_x = 0.483; b0 = -0.022; n=696
power.eq <-function(b1,para){
  p_x = para[1]
  b0=para[2]
  n=para[3]
  Ib1 = (1/(1-p_x))*((1+exp(b0))^2/(1+exp(b0))) +
  (1/p_x)*((1+exp(b0+b1))^2/(1+exp(b0+b1)))
  lambda_weq = n*(b1^2)/Ib1-7.85
}
find para<-multroot(power.eq , start =-0.1, para = c(0.483,-0.022,696), positive = FALSE)
```

- We let power = 90%, MDE  $\beta^* = -0.499$ , then  $RR = \exp(\beta^*) = 0.607$
- Using PASS module: logistic regression with one binary X(Wald test), we can obtain MDE by fixing  $n$ , and try different values of  $RR$  to solve power such that power taking value of 80% or 90% . For example:
  - We first let  $n = 696$ ,  $p_x = 0.483$ , and try  $RR = 0.7$ , then the power = 64.4%

Numeric Results for Two-Sided Wald Test  
Alternative Hypothesis:  $OR_{yx} \neq 1$

| Power   | N   | Percent<br>X=1 | P0     | P1      | ORyx | Alpha | Beta    |
|---------|-----|----------------|--------|---------|------|-------|---------|
| 0.64406 | 696 | 48.3           | 0.4945 | 0.40645 | 0.7  | 0.05  | 0.35594 |

# MDE Calculation: Wald Test

- Then we let try  $RR = 0.650$ , then the power = 80%

Numeric Results for Two-Sided Wald Test

Alternative Hypothesis:  $OR_{yx} \neq 1$

| Power   | N   | Percent<br>X=1 | P0     | P1     | ORyx | Alpha | Beta    |
|---------|-----|----------------|--------|--------|------|-------|---------|
| 0.80011 | 696 | 48.3           | 0.4945 | 0.3887 | 0.65 | 0.05  | 0.19989 |

- When we let  $RR = 0.607$ , then the power = 89.85%

Numeric Results for Two-Sided Wald Test

Alternative Hypothesis:  $OR_{yx} \neq 1$

| Power   | N   | Percent<br>X=1 | P0     | P1      | ORyx  | Alpha | Beta    |
|---------|-----|----------------|--------|---------|-------|-------|---------|
| 0.89851 | 696 | 48.3           | 0.4945 | 0.37257 | 0.607 | 0.05  | 0.10149 |

- When we let  $RR = 0.606$ , then the power = 90.03%

Numeric Results for Two-Sided Wald Test

Alternative Hypothesis:  $OR_{yx} \neq 1$

| Power   | N   | Percent<br>X=1 | P0     | P1      | ORyx  | Alpha | Beta    |
|---------|-----|----------------|--------|---------|-------|-------|---------|
| 0.90033 | 696 | 48.3           | 0.4945 | 0.37218 | 0.606 | 0.05  | 0.09967 |

- Base on our attempts, we can conclude that, when we let  $RR = 0.650$ , the power=80%; when we let  $RR$  between 0.606 and 0.607, the power =90%.

# MDE Calculation: Score Test

- We let  $\text{power} = 80\%$ ,  $n = 696$ ,  $p_x = 0.483$ ,  $\beta_0 = -0.022$ , then MDE  $\beta^* = -0.427$  and  $RR = \exp(\beta^*) = 0.652$ . The formula for calculating MDE is in page 51.  $p_{y1}$  and  $p_{y0}$  in the code are  $p_1$  and  $p_0$ , respectively.

```
library(rootSolve)
p_x = 0.483; b00=-0.173; b0a = -0.022; n=696;

power.eq1 <-function(b1,para){
  b00 = para[1]; b0a=para[2]; n=para[3]; p_x= para[4]
  p_y0 = exp(b0a)/(1+exp(b0a))
  p_y1 = exp(b0a+b1)/(1+exp(b0a+b1))

  lambda_0= n*(p_x*(1-p_x)*(p_y1-p_y0)^2)/
    (p_x*p_y0*(1-p_y0)+ (1-p_x)*p_y1*(1-p_y1))-7.85

  # power = power
}
find.para<-multiroot(power.eq1, start =-0.1,
  para = c(-0.173,-0.022,696,0.483), positive = FALSE)
```

- We let  $\text{power} = 90\%$ , MDE  $\beta^* = -0.495$ , then  $RR = \exp(\beta^*) = 0.610$

# MDE Calculation: LRT Test

- power = 80%,  $p_x = 0.483$ ,  $\beta_{0|H_A} = -0.022$ ,  $\tilde{\beta}_0 = -0.3462$ ,  $n = 696$ , then  $\beta^* = -0.429$ ,  $RR = \exp(\beta^*) = 0.651$ , the formula for calculating MDE is in page 60.
- We let power = 90%, MDE  $\beta^* = -0.498$ , then  $RR = \exp(\beta^*) = 0.608$

```
library(rootSolve)
p_x = 0.483; b0a = -0.022; n=696
power.eql <-function(b1,para){
  b0a=para[1]; n=para[2]; p_x= para[3]
  p0 =exp(b0a)/(1+exp(b0a)); p1 = exp(b0a+b1)/(1+exp(b0a+b1))
  p_star = (1-p_x)*p0+p_x*p1; b_tilde = log(p_star / (1-p_star ))
  D_star = 2*( p_x*(p1*(b0a+b1)-p_star*b_tilde -log( (1+exp(b0a+b1))/(1+exp(b_tilde)) ) )
    +(1-p_x)* (p0*b0a-p_star*b_tilde- log((1+exp(b0a))/(1+exp(b_tilde)) ) ) )
  M_t = (p_x*(1-p1)*p1 +(1-p_x)*p0*(1-p0))/(p_star *(1-p_star ) )
  lambda_leq = n*D_star+1-M_t -7.85
}
find para<-multiroot(power.eql, start =-0.1,
  para = c(b0a,n,p_x), positive = FALSE)
find para$root
[1] -0.4289922
exp(-0.4289922)
[1] 0.651165
```

# Compare three tests

Table 2: Power Analysis Results of HPTN 037 data

|             |             | Wald Test | Score Test | LRT    |
|-------------|-------------|-----------|------------|--------|
| Power       | RR=0.5      | 99.30%    | 99.47%     | 99.38% |
|             | RR=0.75     | 47.01%    | 47.30%     | 47.18% |
| Sample Size | RR=0.5      |           |            |        |
|             | Power =80%  | 280       | 268        | 274    |
|             | Power = 90% | 375       | 359        | 367    |
|             | RR=0.75     |           |            |        |
|             | Power =80%  | 1539      | 1527       | 1533   |
|             | Power = 90% | 2058      | 2043       | 2051   |
| MDE         | Power =80%  | 0.650     | 0.652      | 0.651  |
|             | Power = 90% | 0.607     | 0.610      | 0.608  |

# List of Notation

$i$ :  $i$ th group,  $i=1,2$

$j$ :  $j$ th observation

$\beta^*$ : the parameter of interest

$\beta_0$ : nuisance parameter

$\beta = (\beta_0, \beta^*)^T$ : the vector of parameters

$q$ : number of parameter in  $\beta$

$\beta_{H_0} = (\beta_{0|H_0}, \beta_{0|H_0}^*)^T$ : the vector of parameters under  $H_0$

$\beta_{H_A} = (\beta_{0|H_A}, \beta_{0|H_A}^*)^T$ : the vector of parameters under  $H_A$

$\mu_1$  and  $\mu_2$ : two sample means

$n$ : total sample size

$\mathbf{Y} = (Y_1, \dots, Y_n)^T$ : the outcomes for participants  $j = 1, \dots, n$

$\mathbf{X} = (X_1, \dots, X_n)^T$ : the intervention/ treatment status for participants

$Z^2(\mathbf{Y}, \mathbf{X}; \beta)$ : test statistic which is a function of the data  $\{\mathbf{Y}, \mathbf{X}\}$ .

$\lambda$ : non-central parameter of  $\chi^2$  distribution

$\chi_1^2$ : central  $\chi^2$  distribution with  $DF = 1$

$\chi_1^2(\lambda)$ : non-central  $\chi^2$  distribution with  $DF = 1$  and  $\lambda$

$p_j = \frac{\exp(\beta_0 + \beta^* X_1)}{1 + \exp(\beta_0 + \beta^* X_1)}$ : the probability of  $P(Y_j = 1 | X_j; \beta)$  under logistic regression model

# List of Notation

$p_0$ :  $Pr(Y = 1|X = 0)$  under  $H_A$

$p_1$ :  $Pr(Y = 1|X = 1)$  under  $H_A$

$p^*$ :  $Pr(Y = 1)$  under  $H_A$

$U_j$ : first derivative of the log-likelihood for single observation

$p_x$ : given  $X \sim \text{Bern}(p_x)$ ,  $p_x = Pr(X = 1)$

$\mathbf{U}$ :  $\sum_j^n U_j(Y_j, X_j; \beta)$

$\mathbf{I}$ :  $\mathbf{I} = \text{Var}(\mathbf{U})$ , also be the negative expectation of information matrix of all observations

$I_j(\beta)$ : the negative expectation of information matrix for a single observation

$I_j^{-1}(\beta^*)$ : the  $r - r$ th element of the covariance matrix of  $\text{Var}(\beta)$  based on the  $I_j(\beta)$ .

$\pi$ : the statistical power of a test

$\alpha$ : the significant level of a test

$U_{\beta^*}$ : the  $r$ th element in the score function  $\mathbf{U}$

$U_{\beta_0}$ : the element in the score function for  $\beta_0$

# List of Notation

$\hat{V}_{r-r}(\mathbf{Y}, \mathbf{X})$ : the estimated of the variance of  $U_{\beta^*}(\hat{\beta}_{0|H_0}, \beta^*)$

$\xi$ : the expectation of the approximation of  $U_{\beta^*}(\hat{\beta}_{0|H_0}, \beta^*)$  with respect to  $(\mathbf{Y}, \mathbf{X})$  evaluated as  $\beta_{H_0}^*$  and  $\tilde{\beta}_0$ .

$I^*(\beta^*)$ : the variance of the approximation of  $U_{\beta^*}(\hat{\beta}_{0|H_0}, \beta^*)$  with respect to  $(\mathbf{Y}, \mathbf{X})$  evaluated as  $\beta_{H_0}^*$  and  $\tilde{\beta}_0$ .

$M$ : a matrix which trace is an approximation of the expected value of the second term in (11).

$\Delta$ : the expected value of the third term in (11)

$\Delta^*$ :  $\Delta/n$ .

$Z_\lambda^2$ : a random variable following  $\chi_1^2(\lambda)$  distribution with DF=1, and non-centrality parameter  $\lambda$ .